

Token-Level Uncertainty as a Predictor of Hallucination in Large Language Models

Abstract

Large language models (LLMs) routinely generate fluent but factually unsupported text, a failure mode commonly termed hallucination. Because hallucinated content is often indistinguishable from correct content at the surface level, researchers have increasingly turned to the internal probabilistic signals that LLMs produce during decoding — token-level uncertainty — as a lightweight, reference-free predictor of unreliable generation. This paper surveys and synthesizes the growing body of work on token-level uncertainty quantification (UQ) as a hallucination predictor. We review the theoretical basis for the uncertainty–hallucination relationship, distinguish token-level approaches from semantic-, consistency-, and internal-state-based alternatives, and examine influential methods including sequence log-probability, predictive entropy, Claim Conditioned Probability, focus-weighted token scoring, and attention-based recurrent uncertainty quantification. We then discuss current empirical developments, including scalable and supervised token-level detectors, hidden-state dispersion methods, and hybrid entropy–calibration frameworks. We critically examine the limitations of token-level uncertainty, notably its blindness to semantic equivalence, its vulnerability to confident hallucination, and its sensitivity to decoding strategy and model calibration. We conclude by identifying open research gaps — including cross-lingual and cross-domain generalization, computational efficiency at scale, and the integration of token-level signals with semantic and mechanistic interpretability methods — and outline directions for future research toward more trustworthy language generation.

Keywords: large language models; hallucination detection; uncertainty quantification; token-level entropy; predictive entropy; semantic entropy; calibration; natural language generation

1. Introduction

The rapid deployment of large language models (LLMs) across knowledge-intensive applications — question answering, summarization, medical decision support, and legal drafting — has exposed a persistent and consequential failure mode: hallucination, the generation of fluent text that is factually incorrect, unsupported by evidence, or internally inconsistent with the source input (Ji et al., 2023). Hallucinations are especially dangerous because they are rarely accompanied by any surface marker of unreliability; a fabricated citation or an invented statistic can be stated with the same syntactic confidence as a verified fact. This has motivated a substantial research effort into automatic hallucination detection methods that do not depend on human review or external knowledge bases at inference time.

Among the various detection paradigms — retrieval-based fact verification, multi-sample consistency checking, and internal-state probing — uncertainty quantification (UQ) has emerged as a particularly attractive strategy because it can, in principle, be computed directly from signals the model already produces during generation (Kadavath et al., 2022; Kuhn et al., 2023). The central hypothesis underlying UQ-based detection is that when a model is “guessing” rather than retrieving a well-supported answer from its parametric knowledge, its predictive distribution over candidate continuations will be flatter, and the probability mass assigned to the tokens it eventually emits will be correspondingly lower (Farquhar et al., 2024; Malinin & Gales, 2021). Token-level uncertainty operationalizes this hypothesis at the finest possible granularity: the probability, entropy, or related statistic computed over the model’s output distribution at each individual decoding step.

Token-level uncertainty is attractive for three practical reasons. First, it is computationally inexpensive relative to sampling-based consistency methods such as SelfCheckGPT, which require generating and comparing multiple full responses (Manakul et al., 2023). Second, it is available in “white-box” deployment settings without any additional model calls, since log-probabilities are a byproduct of standard autoregressive decoding. Third, because it operates at the token rather than the sequence level, it offers the possibility of localizing hallucinated spans within an otherwise correct passage, which is valuable for downstream applications such as selective abstention, highlighting, or citation verification (Fadeeva et al., 2024).

At the same time, a growing body of empirical and theoretical work has shown that the relationship between token-level uncertainty and hallucination is neither simple nor uniformly reliable. Confident hallucinations — outputs generated with low predictive entropy despite being factually wrong — pose a fundamental challenge to any confidence-based detector (Kalai & Vempala, 2023). Moreover, naive token probability conflates two distinct sources of uncertainty: uncertainty over which fact to express (semantic uncertainty) and uncertainty over which words to use to express it (lexical uncertainty), a distinction first formalized by Kuhn et al. (2023) in their semantic entropy framework.

This paper provides a structured review of token-level uncertainty as a predictor of hallucination in LLMs. Section 2 situates token-level UQ within the broader hallucination-detection literature and reviews its theoretical foundations. Section 3 provides a technical treatment of the principal token-level uncertainty metrics and the mechanisms by which they are hypothesized to relate to hallucination. Section 4 surveys current approaches and empirical developments, including recent (2024–2026) supervised, attention-based, and dynamics-aware detectors. Section 5 critically examines the challenges and limitations of the paradigm. Section 6 identifies open research gaps and proposes directions for future work, and Section 7 concludes.

2. Background, Related Work, and Literature Review

2.1 Defining Hallucination in LLMs

Hallucination has been defined variously across the natural language generation (NLG) literature, but converges on the notion of generated content that is unfaithful to a source (intrinsic hallucination) or unsupported by any verifiable source, including the model's training data (extrinsic hallucination) (Ji et al., 2023). Zhang et al. (2023) further distinguish factuality hallucinations, which contradict verifiable world knowledge, from faithfulness hallucinations, which diverge from a given input context, such as in summarization or retrieval-augmented generation. Token-level uncertainty methods have primarily targeted factuality hallucination in open-domain generation and question answering, though extensions to faithfulness settings exist (Fadeeva et al., 2024).

2.2 The Uncertainty–Hallucination Hypothesis

The theoretical motivation for uncertainty-based detection draws on a broader literature on predictive uncertainty in deep learning, including calibration (Guo et al., 2017) and deep ensemble methods (Lakshminarayanan et al., 2017), adapted to the autoregressive, structured-prediction setting of language generation (Malinin & Gales, 2021). Kadavath et al. (2022) provided early evidence that LLMs possess some degree of internal self-knowledge: models trained at sufficient scale can produce calibrated estimates of whether their own answers to factual questions are correct, particularly when probability is aggregated appropriately across multiple candidate answers. This finding underpins the more specific claim that token-level generation probabilities, aggregated over a response, carry a meaningful signal about the correctness of that response.

However, Kalai and Vempala (2023) offer a formal argument that calibrated language models are statistically compelled to hallucinate on facts that appear rarely in training data (the “singleton” rate), implying that low-probability generation is not merely an artifact of poor training but a structural consequence of calibrated next-token prediction under data sparsity. This result is important context for token-level UQ research: it suggests that uncertainty signals are most informative precisely in the regime — rare or long-tail facts — where they are hardest to calibrate reliably.

2.3 Taxonomy of Hallucination Detection Approaches

Recent surveys classify UQ-based hallucination detection into several overlapping families (Fadeeva et al., 2024; Ji et al., 2023): (i) token probability-based methods, which estimate uncertainty directly from the output token distribution, including minimum, average, or entropy-weighted token probabilities; (ii) output consistency-based methods, such as SelfCheckGPT, which sample multiple generations and assess semantic or lexical agreement among them (Manakul et al., 2023); (iii) internal-state examination methods, which probe hidden activations, attention patterns, or layer-wise representations for signals correlated with factual error (Vazhentsev et al., 2026; Srey et al., 2026); and (iv) self-checking or verbalized-confidence methods, in which the model is prompted to introspect on the reliability of its own output. Token-level uncertainty, the focus of this review, falls principally in the first category but

increasingly intersects with the third, as several recent methods fuse token probabilities with internal attention or hidden-state signals (Vazhentsev et al., 2026).

2.4 From Token Probability to Semantic Entropy

A pivotal development in this literature is the recognition that raw token-level probability conflates two independent sources of variability. Kuhn et al. (2023) demonstrated that because many distinct token sequences can express the same underlying meaning (“semantic equivalence”), predictive entropy computed directly over token sequences systematically overstates a model’s uncertainty about the fact being expressed. Their proposed semantic entropy clusters sampled generations by bidirectional entailment before computing entropy over the resulting meaning-clusters, and was shown to be more predictive of answer correctness than naive sequence-level token probability across multiple free-form question-answering benchmarks. This method was later validated at scale and published in *Nature*, showing that semantic-entropy-based detectors generalize across model families and domains, including biography generation and open-domain QA (Farquhar et al., 2024). Kossen et al. (2024) subsequently showed that lightweight linear probes trained on hidden states can approximate semantic entropy at a fraction of the computational cost, suggesting that the semantic signal, while distinct from raw token probability, is nonetheless partially recoverable from internal token-level representations.

3. Token-Level Uncertainty: Technical Foundations

3.1 Core Metrics

Given an autoregressive LLM that produces a probability distribution $p(y_t | y_{<t}, x)$ over the vocabulary at each decoding step t , several token-level statistics have been proposed as uncertainty proxies. Sequence log-probability aggregates the log-probabilities of the emitted tokens, typically length-normalized to avoid penalizing longer sequences; Guerreiro et al. (as reviewed in the comprehensive hallucination survey of the arXiv literature) applied this measure to detect hallucinations in machine translation by flagging outputs whose normalized log-probability falls below a threshold. Predictive (Shannon) entropy over the full vocabulary distribution at each step provides a complementary measure that captures the shape of the distribution rather than only the probability of the selected token, and can be aggregated across a sequence via summation, averaging, or maximum-entropy pooling (Malinin & Gales, 2021).

A key refinement is Claim Conditioned Probability (CCP), introduced by Fadeeva et al. (2024), which decomposes token-level uncertainty about an atomic claim into two components — uncertainty about which claim to generate at a given step, and uncertainty about which surface form to use to express a chosen claim — and removes the latter, claim-irrelevant component. This addresses, at the token level, the same lexical-variation problem that motivated semantic entropy at the sequence level. Human evaluation showed that a fact-checking pipeline built on

CCP was competitive with a retrieval-based fact-checking tool, without requiring external knowledge access, across seven LLMs and four languages (Fadeeva et al., 2024).

A separate line of work improves token-level uncertainty by reweighting tokens according to their informativeness. Zhang et al. (2023) propose a reference-free method that mimics human fact-checking behavior along three axes: attending to the most informative keywords in the generated text (rather than treating all tokens equally), discounting the contribution of tokens whose historical context was itself unreliable (to avoid double-penalizing cascading errors), and adjusting for token properties such as part-of-speech and frequency, since function words and high-frequency tokens are near-deterministic and therefore uninformative about hallucination regardless of their raw probability.

3.2 Internal-State and Attention-Based Extensions

More recent work extends token-level uncertainty beyond output-layer probabilities to intermediate model computations. Vazhentsev et al. (2026) introduce Recurrent Attention-based Uncertainty Quantification (RAUQ), based on the empirical observation that specific “uncertainty-aware” attention heads systematically reduce their focus on preceding context tokens when the model is about to generate incorrect information. RAUQ automatically identifies these heads and fuses their activation patterns with token-level confidence scores in a recurrent scheme, producing a sequence-level uncertainty estimate in a single forward pass with under one percent additional computational overhead, and was shown to outperform prior unsupervised UQ baselines across twelve datasets spanning question answering, summarization, and translation, and nine LLMs. Srey et al. (2026) similarly move beyond last-token or mean-pooled representations, proposing Sequential Internal Variance Representation (SIVR), which aggregates token-wise, layer-wise dispersion of hidden states across the full generated sequence to capture temporal patterns indicative of factual error, avoiding the strong assumptions about monotonic hidden-state evolution made by some earlier internal-state probes.

3.3 Why Token-Level Signals Should (and Should Not) Predict Hallucination

The mechanistic justification for the uncertainty–hallucination link rests on the premise that a model’s next-token distribution reflects the degree to which training data supports a given continuation: well-attested facts produce sharply peaked distributions, while facts that are rare, contradictory across sources, or absent from training data produce flatter distributions, forcing the model to “guess” among several plausible completions (Farquhar et al., 2024). Under this account, token-level uncertainty is not merely correlated with hallucination but is, in expectation, a direct symptom of the underlying condition — insufficient evidence — that causes it.

This account, however, is incomplete in at least three respects, each of which motivates the limitations discussed in Section 5. First, uncertainty about phrasing is conflated with uncertainty about content unless explicitly disentangled, as in semantic entropy or CCP (Kuhn et al., 2023; Fadeeva et al., 2024). Second, the premise assumes that models are well-calibrated, i.e., that

predictive probability tracks empirical correctness frequency; post-training procedures such as instruction tuning and reinforcement learning from human feedback are known to reduce calibration quality relative to base pretrained models, which can systematically distort the uncertainty–correctness relationship (Guo et al., 2017; Kadavath et al., 2022). Third, and most fundamentally, a model can be sharply confident about a memorized but incorrect association, in which case low entropy will co-occur with hallucination rather than predict its absence — the “confident hallucination” phenomenon discussed further in Section 5.

4. Current Approaches and Developments

4.1 Supervised and Scalable Token-Level Detectors

While early token-level UQ methods were largely unsupervised, recent work trains dedicated detectors on token-level hallucination annotations. Min et al. (2026) introduce TokenHD, a pipeline combining a scalable synthetic data engine for generating token-level hallucination labels with an importance-weighted training recipe, arguing that step-level detection methods such as process reward models suffer from limited granularity and dependence on explicit step segmentation. Their detector, trained on free-form text without predefined segmentation, was shown to scale consistently in performance from 0.6B to 8B parameters and to surpass much larger reasoning models such as QwQ-32B on reasoning-intensive hallucination detection, indicating that token-level supervision can substitute for raw model scale under a well-designed training procedure.

4.2 Denoising-Dynamics and Non-Autoregressive Extensions

As diffusion-based (non-autoregressive) large language models have emerged as an alternative generation paradigm, token-level uncertainty has been adapted accordingly. DynHD (Qian et al., 2026) observes that in the fixed-length, iterative-refinement generation process of diffusion LLMs, tokens contribute unevenly to hallucination signals, and that the temporal evolution of token uncertainty across the denoising trajectory itself carries information. The method combines a semantic-aware evidence construction module, which filters non-informative tokens, with a deviation-based detector that compares the observed uncertainty trajectory to a learned reference trajectory, illustrating that the token-level uncertainty paradigm generalizes beyond the autoregressive setting in which it was originally developed.

4.3 Calibration-Aware and Hybrid Frameworks

A further development addresses the confident-hallucination blind spot directly by combining entropy-based measures with calibration diagnostics. Recent work has argued that traditional token- and semantic-entropy measures fail precisely when models produce high-confidence hallucinations — consistently generating the same incorrect output with a sharply peaked probability distribution — and has proposed incorporating expected calibration error (ECE) and verbalized (self-declared) confidence as complementary signals that can catch systematic

overconfidence that entropy alone cannot detect. Such hybrid frameworks treat token-level uncertainty as one input among several — alongside consistency-based and self-assessment signals — rather than a standalone detector, reflecting a broader trend toward ensemble UQ pipelines (see also the classification schema of token-, semantic-, and structure-based methods in recent comprehensive surveys of LLM hallucination).

4.4 Comparative Summary of Representative Methods

Method	Signal	Key Mechanism	Source
Sequence log-probability	Token-level output probability	Length-normalized aggregation of emitted-token log-probabilities	Malinin & Gales (2021)
Semantic entropy	Meaning-clustered sequence entropy	Clusters sampled generations by bidirectional entailment before computing entropy	Kuhn et al. (2023); Farquhar et al. (2024)
Claim Conditioned Probability	Token-level, claim-decomposed probability	Removes surface-form (lexical) variance from claim-level uncertainty	Fadeeva et al. (2024)
Focus-weighted uncertainty	Reweighted token probability	Weights tokens by informativeness, historical reliability, and token properties	Zhang et al. (2023)
Semantic entropy probes	Linear probe on hidden states	Approximates semantic entropy cheaply from a single hidden state	Kossen et al. (2024)
RAUQ	Attention-weighted recurrent token confidence	Identifies “uncertainty-aware” attention heads and fuses with token confidence	Vazhentsev et al. (2026)
SIVR	Layer-wise hidden-state dispersion	Aggregates per-token variance across layers and sequence positions	Srey et al. (2026)
TokenHD	Supervised token-level classifier	Trains on synthetically labeled token-level hallucination data	Min et al. (2026)

Table 1. Representative token-level and closely related uncertainty-quantification methods for hallucination detection.

5. Research Challenges and Limitations

Confident hallucination. Perhaps the most consequential limitation of token-level uncertainty is its failure to detect hallucinations that are generated with high model confidence. When a model has memorized a spurious but statistically sharp association — for instance, a plausible-sounding but incorrect numeric fact repeated across low-quality training sources — its predictive distribution can be sharply peaked around the incorrect token, yielding low entropy and high token probability despite factual error. This decouples the theoretical premise of uncertainty-based detection (low confidence signals insufficient evidence) from its practical

failure mode (confident generation of unsupported content), and has been identified as a critical blind spot for entropy-based detectors that calibration-aware metrics such as expected calibration error attempt, only partially, to address.

Semantic–lexical conflation. As established by Kuhn et al. (2023), naive token-level uncertainty measures cannot distinguish uncertainty over meaning from uncertainty over wording. A model that is fully confident about a fact but has many equally plausible ways of phrasing it will exhibit artificially high raw entropy, producing false positives. While semantic entropy and Claim Conditioned Probability were designed specifically to address this conflation, both require either multiple sampled generations (increasing inference cost) or claim-level decomposition (increasing pipeline complexity), illustrating a persistent efficiency–fidelity trade-off in the field.

Calibration drift under alignment. Instruction tuning and reinforcement learning from human feedback, now standard in deployed LLMs, are known to alter the correspondence between output probability and empirical accuracy relative to base pretrained models (Guo et al., 2017; Kadavath et al., 2022). Because token-level UQ methods implicitly assume some degree of calibration, their reliability may vary substantially across the base-model-to-aligned-model pipeline, and results validated on one stage of training may not transfer to another.

Structural inevitability of low-probability generation. Kalai and Vempala (2023) show that even a perfectly calibrated language model must, in expectation, hallucinate on facts that appear as singletons in its training corpus, since calibration requires assigning non-negligible probability mass to plausible-but-unverified completions when no single correct answer dominates the training distribution. This result implies a theoretical ceiling on how well any token-level probability threshold can simultaneously minimize hallucination and preserve coverage of rare-but-true facts, since both are statistically entangled with low training-data frequency.

Threshold and dataset dependence. Empirical performance of token-level detectors is sensitive to decoding strategy (e.g., greedy versus nucleus sampling), sequence length normalization scheme, and the domain and language of evaluation. Fadeeva et al. (2024) note that their CCP-based method required evaluation across four languages and multiple LLMs to establish generality, underscoring that single-language, single-model validation — still common in the literature — risks overstating generalizability.

Localization versus attribution. Although token-level granularity in principle allows the localization of hallucinated spans within a passage, cascading errors mean that a single unreliable early token can propagate uncertainty (or spuriously suppress it) through subsequent generation. Zhang et al. (2023) explicitly address this by discounting tokens generated in unreliable historical context, but the general problem of separating an error's origin from its downstream symptoms remains only partially solved, particularly in long, multi-hop reasoning chains where token-level detectors underperform relative to structured step-level or process-reward approaches (Min et al., 2026).

Computational and deployment constraints. Methods requiring access to full output-layer probability distributions or internal hidden states (“white-box” access) are inapplicable to hallucination detection over commercial API-only models, which typically expose limited or no token-level probability information. This restricts token-level UQ largely to open-weight model deployments, motivating black-box alternatives such as consistency sampling (Manakul et al., 2023) or geometric embedding-based approaches, which trade computational cost for applicability.

6. Research Gaps and Future Directions

Robustness to confident hallucination remains the most pressing open problem. Future work could combine token-level uncertainty with orthogonal signals — such as retrieval-based verification, cross-model disagreement, or mechanistic probes for “truthfulness” directions in representation space — in principled ensembles rather than ad hoc pipelines, with an emphasis on characterizing precisely when token-level signals are expected to fail rather than treating failures as unpredictable noise.

Efficient semantic disentanglement. Current methods for separating semantic from lexical uncertainty either require multiple stochastic samples (Kuhn et al., 2023; Farquhar et al., 2024) or specialized claim decomposition (Fadeeva et al., 2024), both adding inference-time cost. The demonstration by Kossen et al. (2024) that semantic entropy can be approximated by a lightweight linear probe on hidden states suggests a promising direction: learning compact, single-pass approximations of semantically disentangled uncertainty that preserve most of the detection benefit of multi-sample methods at a fraction of the cost, and extending this idea specifically to token-level (rather than only sequence-level) granularity.

Cross-lingual and cross-domain generalization. Most token-level UQ evaluations remain concentrated on English-language, general-domain question answering and biography generation. Given evidence that uncertainty calibration varies with language and domain (Fadeeva et al., 2024), systematic benchmarking across low-resource languages, specialized domains such as clinical or legal text, and multi-hop reasoning tasks is needed to establish the boundaries of the paradigm's applicability.

Theoretical characterization of the uncertainty–calibration–alignment interaction. Although it is empirically known that instruction tuning and RLHF affect model calibration, the field lacks a systematic theoretical or large-scale empirical account of how specific alignment procedures reshape the token-level uncertainty–hallucination relationship. Establishing this relationship would allow practitioners to select or recalibrate uncertainty thresholds appropriately for a given post-training regime rather than relying on empirically tuned, model-specific thresholds.

Extension beyond autoregressive decoding. The emergence of diffusion-based and other non-autoregressive LLM architectures, and the corresponding adaptation of token-level uncertainty to denoising dynamics (Qian et al., 2026), suggests a broader need to re-derive

token-level uncertainty theory for generation paradigms where the notion of a strictly sequential “token probability” does not directly apply, including speculative decoding and parallel-sampling architectures.

Black-box and API-constrained detection. As commercial deployment increasingly restricts access to raw token probabilities, future work should prioritize methods that approximate token-level uncertainty signals from limited API outputs (e.g., top-k log-probabilities, if exposed, or repeated low-temperature sampling), bridging the current gap between white-box research settings and real-world, API-mediated deployment constraints.

Standardized, decision-theoretic evaluation. Much of the literature reports detection performance using AUROC or similar ranking metrics computed against post-hoc human or automated correctness labels. Future work would benefit from evaluation protocols grounded in the downstream decision the uncertainty signal is meant to support — for example, selective abstention with an explicit cost model for false negatives (undetected hallucination) versus false positives (unnecessary abstention on correct content) — to make cross-method comparisons more directly actionable for system designers.

7. Conclusion

Token-level uncertainty occupies a distinctive position in the hallucination-detection literature: it is inexpensive, requires no external knowledge source, and is available as a byproduct of standard decoding, making it an attractive first line of defense against unreliable generation. The theoretical grounding for its use — that low predictive confidence reflects insufficient evidentiary support in training data — is well supported by empirical work linking token- and sequence-level probability to answer correctness (Kadavath et al., 2022; Farquhar et al., 2024). At the same time, this review has shown that naive token-level probability is an incomplete predictor: it conflates lexical and semantic uncertainty, is structurally undermined by the phenomenon of confident hallucination, and is sensitive to model calibration shifts induced by alignment procedures.

The trajectory of recent research — from raw sequence log-probability, through semantic entropy and Claim Conditioned Probability, to attention-aware and dynamics-aware detectors, and finally to supervised, scalable token-level classifiers — reflects a field progressively refining what “uncertainty” should mean in the context of factual reliability. The most promising current systems no longer treat token-level probability in isolation but fuse it with semantic clustering, internal attention signals, hidden-state dispersion, or calibration diagnostics. Continued progress will likely depend less on the discovery of a single superior uncertainty statistic and more on principled integration of token-level signals with complementary sources of evidence, together with rigorous, decision-relevant evaluation across languages, domains, and deployment constraints. As LLMs are further integrated into high-stakes decision pipelines, the reliability of

these uncertainty-based safeguards will remain a central, unresolved, and consequential research problem.

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